OV-TCS: Measuring Temporal Label Consistency in Streaming Open-Vocabulary 3D Perception

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Abstract

Streaming open-vocabulary 3D perception systems label 3D instances online, frame by frame — yet the field evaluates them with per-frame detection metrics (mAP/NDS) that cannot see whether an object keeps a consistent identity and label over time, and classical tracking metrics require ground-truth track identities and a closed vocabulary, neither of which this setting provides. We introduce OV-TCS, a temporal label-consistency score computed purely from a system's own tracks and predicted labels, $\text{OV-TCS} = L_{\text{norm}} \cdot (1 - \text{CSR})$, which penalizes both track fragmentation and semantic flicker. We validate the metric rather than assert it: OV-TCS degrades monotonically under controlled fragmentation that provably leaves mAP constant; it predicts label correctness beyond track length on 6,202 ScanNet200 instances; and — without using any ground truth — it ranks systems consistently with HOTA, IDF1, and AMOTA on nuScenes (per-scene r up to 0.87 with HOTA), with rankings invariant across a 30-run sweep of associator hyperparameters. The practical payoff: OV-TCS separates real design decisions that mAP scores as exact ties. Switching from ego- to global-frame association leaves mAP bit-identical while improving OV-TCS by 38% ($0.136 \rightarrow 0.188$; pooled gain +0.052, 95% CI $[0.049, 0.055]$) and reducing fragmentation by 58%. The gain survives the metric's own anti-gaming audit — including a GT-instance-paired comparison (+0.147, 95% CI $[0.142, 0.152]$ over 8,046 instances) — and it replicates on a fully open-vocabulary detection stream (+26%). Code and evaluation protocols will be released.

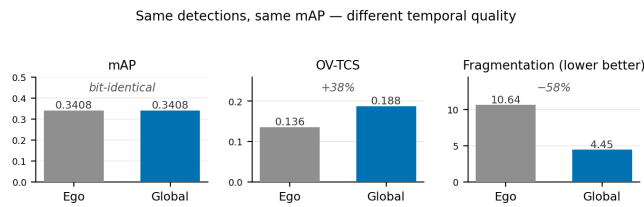

Figure 1: **Why a temporal axis is needed.** Two real systems that differ only in their association frame (ego vs. global) are *bit-identical* in mAP — a practitioner comparing them by detection metrics sees no difference at all — yet differ substantially in temporal label quality: OV-TCS +38% (pooled gain +0.052, 95% CI $[0.049, 0.055]$, 96.7% of scenes improve) and 58% less fragmentation. nuScenes val150, frozen proposals, cache replay; values from Tab. 4.

1 Introduction

Open-vocabulary 3D perception has moved rapidly from offline, per-scene labeling [1, 3, 4] toward *streaming* systems that build semantic 3D maps online for embodied agents [5]. In the streaming setting, the same physical object is observed in many frames, and the system must maintain both its identity (via data association) and its open-vocabulary label over time. A semantic map whose labels flicker between frames, or whose objects shatter into short-lived fragments, is of limited use downstream regardless of how accurate each individual frame is.

Yet the community evaluates these systems almost exclusively with per-frame detection metrics — mAP and, on nuScenes, NDS [9]. These metrics score each frame independently and are structurally blind to cross-frame behavior. We show this blindness is not hypothetical: on nuScenes, replacing the associator's ego-frame matching with global-frame matching leaves mAP and NDS *bit-identical* (association adds or removes no boxes at this operating point), while every temporal quality we can measure changes substantially (Fig. 1, Sec. 6). A practitioner choosing between these two systems by mAP would see no difference at all.

The natural alternative — MOT metrics such as MOTA [12], IDF1 [13], HOTA [14], or nuScenes AMOTA [10] — does not transfer to this setting: all of them require ground-truth track identities and a fixed, closed category set. Open-vocabulary streaming systems produce labels outside any annotated taxonomy, and most target environments (e.g., ScanNet-style indoor scans) have no GT track annotations at all. What is missing is a metric that measures temporal label consistency *intrinsically*: from the system's own tracks and labels, without GT identities, and without restricting the vocabulary.

We propose **OV-TCS** (Open-Vocabulary Temporal Consistency Score). For each track produced by the system's associator, we take the per-frame argmax open-vocabulary labels along the track and score the track by OV-TCS $= L_{\text{norm}} \cdot (1 - \text{CSR})$, where $L_{\text{norm}} = 1 - 1/L$ measures track integrity (fragmentation axis) and $1 - \text{CSR}$, with CSR the fraction of adjacent-frame label switches, measures semantic stability (flicker axis).

Contributions. This paper makes one primary and one secondary contribution. The *primary* contribution is OV-TCS itself: a GT-free, vocabulary-free temporal label-consistency metric, together with the experimental evidence that it is a valid measurement instrument. The formula is deliberately simple; what a metric paper owes the reader is not a complex formula but proof that the simple one measures what it claims, and we treat "is this a valid metric?" as an experimental question. The *secondary* contribution is a class-aware 2D→3D label-fusion scheme that improves detection quality while provably leaving track topology unchanged (Sec. 6.3) — a method-level result that also serves as a worked example of the two evaluation axes moving independently.

Validation protocol. The five properties below are not five contributions; they are the validation protocol for the primary contribution — the experimental case, assembled step by step, that OV-TCS is a valid instrument:

(1) It measures what mAP cannot. Injecting track fragmentation *after* detection leaves mAP fixed by construction and measurement, while OV-TCS degrades monotonically; interrupting association (reducing track max-age) degrades both — a positive control (Sec. 5.1).

(2) It is not track length renamed. On ScanNet200 (6,202 instances), OV-TCS predicts per-instance label correctness after controlling for track length (ΔR^2=0.014, F=89.9, p=3.5×10^{-21}); track length alone is nearly useless (partial $r = -0.05$) (Sec. 5.2).

(3) Both factors are necessary, and the product is the minimal correct combination. Flicker moves only $(1 - \text{CSR})$; fragmentation moves only L_{norm} — and pushes $(1 - \text{CSR})$ *up* (shattered tracks look internally stable), so a stability-only metric scores fragmentation as an improvement. Additive combinations collapse onto one factor when tuned (λ^*=0) (Sec. 5.3).

(4) It agrees with GT-based MOT metrics without using GT. On nuScenes val (150 scenes), OV-TCS ranks systems consistently with HOTA, IDF1, and AMOTA, and correlates per-scene with HOTA up to r=0.87 (Sec. 5.4).

(5) Its conclusions are robust to evaluator hyperparameters. Across a 30-run sweep of association gate, max-age, and score threshold, no method ordering flips (Sec. 5.5).

With the metric validated, we use it. It exposes temporal-quality differences between real methods that mAP scores as exact ties (Fig. 1: ego- vs. global-frame association, OV-TCS +38%, fragmentation −58% at bit-identical mAP), and the effect replicates when the label stream is fully open-vocabulary rather than a closed detector's (Sec. 6). The label-fusion contribution moves the opposite axis — mAP 0.3408 → 0.3420 at unchanged track topology — so the two contributions together demonstrate that the evaluation genuinely factorizes into a detection axis and a temporal axis, and that both must be measured (Sec. 6.3).

We are explicit about scope. OV-TCS is validated at *instance* granularity; aggregated per-scene it does not predict per-scene AP (partial $r = -0.03$, p=0.61), and we report this. Absolute OV-TCS values depend on the association operating point and are comparable only at a fixed, stated setting; what is robust is the induced ordering of methods. We view OV-TCS as a necessary complement to mAP for streaming open-vocabulary 3D perception, not a replacement.

2 Related Work

Open-vocabulary 3D perception. Distilling 2D vision–language features into 3D representations [1, 2] and querying 3D instance masks with open vocabularies [3, 4] established the offline setting; ConceptFusion [5] and successors brought open-vocabulary features into online mapping. Evaluation in all of these works is per-frame or per-scene (AP, mIoU); the temporal consistency of the predicted *labels* over a stream is not measured. Our metric targets exactly this gap and is agnostic to how the underlying system produces masks, boxes, or labels.

Tracking metrics. CLEAR-MOT [12] (MOTA/MOTP), IDF1 [13], HOTA [14], and nuScenes AMOTA [10] measure association quality against annotated GT trajectories over a closed category set. Video segmentation has analogous consistency metrics — STQ for segmenting-and-tracking-every-pixel [15], VPQ for video panoptic segmentation [16], and TETA's decomposed tracking evaluation [17] — but all of them likewise require dense GT tube identities over a fixed taxonomy. Open-vocabulary *tracking* benchmarks such as TAO [18] and OVTrack's evaluation protocol [19] broaden the category set, but still score against annotated GT trajectories on a fixed benchmark vocabulary — they extend the taxonomy, not the GT-free setting. They are all the right tools when GT identities exist. OV-TCS is not a competitor in that regime: it is a GT-free, vocabulary-free measurement for settings where those requirements fail. Empirically, it aligns with the association-sensitive components of MOT metrics (HOTA's AssA, IDF1) rather than the detection-sensitive ones (DetA, MOTP) — Sec. 5.4 — which is the alignment one would want from a GT-free surrogate.

Metric validation. HOTA [14] argued that tracking metrics must be decomposable along the failure modes they claim to measure. We adopt the same philosophy for the open-vocabulary streaming setting: each factor of OV-TCS is tied to one controlled corruption axis (flicker, fragmentation), and the combination rule is selected by which forms remain correct on *both* axes, not by which maximizes a single regression score.

3 OV-TCS

3.1 Setting

A streaming system consumes a sensor stream and, at each frame t, outputs a set of localized instances with open-vocabulary labels. An associator links instances across frames into tracks. We make no assumption about the detector, the vocabulary, or the associator beyond this interface: OV-TCS is computed from (i) the track membership the system itself produces and (ii) the per-frame argmax label of each tracked instance. No ground truth of any kind is used to compute the metric.

3.2 Definition

Let a track observe labels $\ell_1, \ldots, \ell_L$ over L frames. Define

$$\mathrm{CSR} = \frac{1}{L-1} \sum_{t=1}^{L-1} \mathbb{1}[\ell_t \neq \ell_{t+1}], \qquad L \geq 2, \tag{1}$$

$$L_{\mathrm{norm}} = 1 - \tfrac{1}{L}, \tag{2}$$

$$\text{OV-TCS} = L_{\mathrm{norm}} \cdot (1 - \mathrm{CSR}). \tag{3}$$

CSR (class-switch rate) measures *semantic flicker*: how often the label changes between adjacent observations. L_{norm} measures *track integrity*: a system that fragments an object into many short tracks populates the evaluation with low-L_{norm} pieces. For a singleton track (L=1), CSR is undefined but OV-TCS is 0 by construction (L_{norm}=0). The system-level score is the mean over *all* tracks: singletons enter at exactly 0, so a system that shatters objects into single-frame fragments is penalized maximally rather than having those fragments silently dropped from the average. We report the per-track distribution, track length, fragmentation (number of predicted tracks matched to one GT instance, used only for *diagnosis* where GT boxes exist), and CSR alongside it (in diagnostic tables, singletons contribute CSR=0 under the same convention).

Both factors are bounded in $[0, 1]$, so OV-TCS $\in [0, 1]$, with 1 attained by an unbroken track that never changes its label. Section 5.3 justifies the product form empirically; here we note only the design intent: a metric for streaming label quality must fail a system for either failure mode, and multiplication is the simplest form with that property.

3.3 What OV-TCS is not

It is not a detection metric (it never inspects localization quality — mAP remains necessary); it is not an MOT metric (it uses no GT identities and therefore cannot detect identity *transfer* between two objects of the same class, a limitation we discuss in Sec. 7); and it is not a per-scene quality score (Sec. 5.2).

3.4 Reporting rule and gameability

Because the system-level score averages over the track population it is computed on, any intervention that filters that population can move the score without improving temporal quality: in our own sweep, raising the proposal-score threshold to 0.3 lifts mean OV-TCS (0.136 → 0.160) while *lowering* mAP (0.341 → 0.310) and cutting the scored population from 341,663 to 73,184 tracks — it simply deletes weak, unstable tracks. Similarly, two systems can differ in population size (Tab. 4: 341,663 vs. 266,623 tracks). We therefore require OV-TCS to be reported jointly with (i) the number of scored tracks, (ii) the detection metric at the same operating point, and (iii) a fixed, stated associator setting. OV-TCS is a companion column to mAP, and comparisons that raise OV-TCS while shrinking the scored population or the detection metric must be treated as suspect until a population-controlled audit shows otherwise.

The rule applied to our own headline. Our flagship comparison (Tab. 4) raises OV-TCS while reducing the track count by 22% (341,663 → 266,623), so our own rule demands an audit: is the gain merging, or filtering? Four population controls answer this (computed on the MOT-comparable 7-class tracking subset: 700,752 boxes; 293,275 vs. 230,644 tracks). *(i) Box conservation:* both arms contain exactly the same boxes, per-sample-equal in all 6,019 samples (1,029,380 boxes bit-identical on the full population) — association regroups boxes but cannot delete them, so no track can be suppressed, only re-grouped. *(ii) Scored coverage rises:* the fraction of boxes covered by multi-frame ($L \geq 2$) tracks grows from 82.4% to 87.3%; fewer tracks here means merging, which *adds* former singletons to the scored population — the opposite of filtering. *(iii) Within-length-stratum gains:* global $\geq$ ego in every stratum from L=2 to L=19 (e.g., L=2: 0.163 vs. 0.138), and reweighting global's per-length means to ego's exact length distribution retains a +14.7% gain with the length distribution frozen. *(iv) GT-instance-paired:* over the 8,046 GT instances matched in both arms (nearest-GT $\leq$ 2 m BEV per frame, majority vote per track), per-instance OV-TCS rises 0.287 → 0.434 (+0.147, 95% CI $[0.142, 0.152]$, 10^4-resample paired bootstrap; 70.9% of instances improve, 11.5% tie) and per-instance fragmentation falls 7.89 → 4.65. On a fixed set of physical objects the gain is *larger* than the system-level delta. The headline gain is merging, not filtering — and this audit is the template

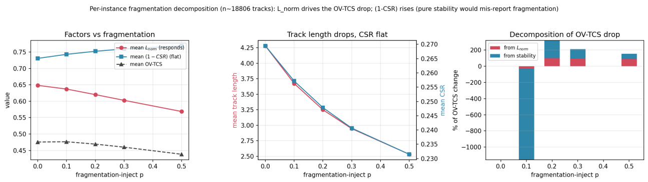

Figure 2: Controlled fragmentation ($p = 0 \to 0.5$, nuScenes val150, ∼20k tracks per level; detections frozen). mAP is constant by construction and measurement. OV-TCS falls monotonically, and the decomposition shows the drop is carried entirely by L_{norm} while the stability factor $(1 - \text{CSR})$ *rises* — the case that breaks stability-only metrics (Sec. 5.3).

we prescribe for any comparison the reporting rule flags.

4 Experimental Setup

Indoor. ScanNet200 [11] validation (312 scenes). Instances are tracked by projecting Mask3D [6] 3D instance masks through the RGB-D stream; per-frame open-vocabulary labels come from a CLIP-based labeler over the 200-class prompt set. This yields 6,202 tracked instances with GT semantic labels, used for predictive validation (does OV-TCS predict whether the track's majority label is correct?).

Outdoor. nuScenes [9] validation, 150 scenes, single sweep. Proposals come from CenterPoint [7] (closed 10-class stream; the "native" anchor, mAP 0.3408/NDS 0.3150) and from a detection-guided open-vocabulary pipeline (YOLO-World [8] 2D detections lifted to 3D clusters; 15-class prompt vocabulary). All temporal experiments are cache replays over frozen proposals, so detection content is held exactly constant while association and labeling vary; mAP/NDS are computed with the official devkit.

Associators. A class-agnostic centroid associator (BEV gate 2.0 m, max-age 5) in either ego frame (no ego-motion compensation; the production configuration) or global frame. All parameters are swept in Sec. 5.5.

5 Validating the Metric

5.1 mAP is blind to temporal degradation

We corrupt track topology while provably leaving detections untouched: with probability p, a track is severed between adjacent frames (fragmentation injection), acting purely on associator output. mAP is constant across all p — confirmed by measurement, not only by construction — while OV-TCS falls monotonically (Fig. 2). On the open-vocabulary stream, fragmentation $0 \to 0.5$ moves mean OV-TCS $0.476 \to 0.438$ (mean over the $L \geq 2$ tracks the injection acts on) while mAP does not move at all. As a positive control, corrupting association *itself* (max-age $5 \to 1$, which interrupts tracks and changes what is emitted over time in a tracking-aware evaluation) degrades both families of temporal metrics; and in the E4 sweep (Sec. 5.5) mAP is bit-identical across every association gate and max-age setting, isolating detection from association exactly. Per-frame detection metrics cannot see the axis OV-TCS measures.

5.2 Beyond track length

A fair worry is that OV-TCS is track length with extra notation. We test whether OV-TCS predicts *label correctness* (majority label of the track equals the GT class of its instance) on 6,202 ScanNet200 instances, controlling for length. OV-TCS retains a significant partial correlation given length (r=0.12, p=3.5×10^{-21}), and adding OV-TCS to a length-only model yields ΔR^2=0.014 (F=89.9). Length alone is nearly useless (partial r=−0.05; $\Delta R^2\approx$0.003). The effect size deserves honest framing: ΔR^2=0.014 is small in absolute terms, and OV-TCS is a coarse topological signal, not a calibrated correctness predictor. The relevant comparison, however, is not against an oracle but against the only other GT-free covariate available — track length — and the direction of the effect is estimated with high reliability (F=89.9, n=6202). The test establishes exactly what the metric needs: information about label correctness beyond length, in the direction it claims. (Sec. 5.3 shows the stability factor alone carries even more of this particular signal — and why that does not make it a better metric.)

The signal lives at instance granularity. Aggregated per scene (n=312) and regressed against per-scene AP_{50}, OV-TCS has no predictive power (partial r= − 0.03, p=0.61). We therefore never use OV-TCS as a scene score, and report it as a per-track distribution or its mean over a fixed large track population.

5.3 Why the product: two failure modes, two factors

Why $L_{\text{norm}} \cdot (1 - \text{CSR})$, and not stability alone, length alone, or a sum? We answer with two independent corruption axes (Tab. 1).

Flicker axis. On real indoor flicker, explanatory power for label correctness is owned by the stability factor: stability-only achieves ΔR^2=0.023, the product 0.014, and length-only 0.003. A validation-tuned weighted sum collapses to stability-only (λ^*=0.00). If flicker were the only failure mode, $(1 - \text{CSR})$ alone would win.

Fragmentation axis. It is not the only failure mode. Under controlled fragmentation $0 \to 0.5$ on nuScenes (∼19–22k tracks per level), mean track length falls $4.28 \to 2.53$ and L_{norm} $0.648 \to 0.569$, but mean $(1-\text{CSR})$ *rises* $0.731 \to 0.769$: shattering a track leaves fewer

Formulation	flicker axis ΔR^2	fragmentation axis
$(1-\mathrm{CSR})$ only (stability)	**0.023**	*misreports* (↑)
$L_{\mathrm{norm}} \cdot (1-\mathrm{CSR})$ (product)	0.014	correct (↓)
$\min(L_{\mathrm{norm}}, 1-\mathrm{CSR})$	0.012	correct
geometric mean	0.011	correct
harmonic mean	0.010	correct
L_{norm} only (length)	0.003	correct (↓)
weighted sum (tuned)	$\lambda^*=0 \Rightarrow$ stability	misreports

Table 1: Formulation analysis. Left: explanatory power for label correctness under real flicker (ScanNet200, n=6202). Right: qualitative behavior under controlled fragmentation (nuScenes, ∼20k tracks per level). Only forms that keep both factors respond correctly on both axes; the product is the simplest of them and the strongest on the flicker axis among them.

within-track transitions, so each fragment looks internally stable. A stability-only metric therefore scores fragmentation as an *improvement*. Decomposing the product's response, the L_{norm} factor carries the entire degradation (contribution −0.058 vs. net −0.037) against a +0.025 opposing contribution from stability.

Conclusion, stated precisely. Each factor exclusively owns one failure mode; dropping either factor makes the metric *directionally wrong* on the other axis, and tunable additive forms empirically collapse to a single factor. Among the two-factor forms that remain correct on both axes (product, min, geometric, harmonic), the product is the simplest and carries the most flicker-axis signal. Two honesty notes bound this claim. First, the fragmentation axis is established by *controlled injection*; on the real method pair of Tab. 4, fragmentation and flicker happen to improve together, so a stability-only score produces the same ranking there — the injected case is precisely the trade-off regime a metric must not get wrong, but we have not yet observed that regime between two deployed methods. Second, we do not claim the product is uniquely optimal; we claim it is the minimal form that is never directionally wrong on either observed failure mode.

5.4 Agreement with GT-based MOT metrics

If OV-TCS measures temporal quality, it should agree with GT-based MOT metrics where those are computable. On nuScenes val150 we evaluate the ego- and global-association arms with the official tracking toolkits (Tab. 2). OV-TCS ranks the arms identically to HOTA, IDF1 and AMOTA, and the identity-switch count falls by 2.8× where OV-TCS improves.

Per-scene (n=150 per arm, Fig. 3), OV-TCS correlates strongly with the association-driven metrics — HOTA r=0.87 in the ego arm, 0.60 in the global arm, 0.78 pooled; AssA 0.77 pooled; IDF1 0.71 pooled — and

Arm	OV-TCS	HOTA	AssA	IDF1	AMOTA	IDS	Frag
Ego	0.136	0.126	0.161	0.087	0.034	87,003	9,376
Global	**0.188**	**0.199**	**0.400**	**0.183**	**0.158**	**30,608**	9,452

Table 2: System-level agreement on nuScenes val150 (cache replay; class-agnostic association, 2.0 m gate). OV-TCS, computed without GT identities, ranks the two systems identically to HOTA, IDF1, and AMOTA (official devkit). Frag here is CLEAR-MOT's interruption count over GT-matched trajectories — a different quantity from the GT-instance fragmentation of Tab. 4, and roughly constant here because both arms match a similar set of GT trajectories.

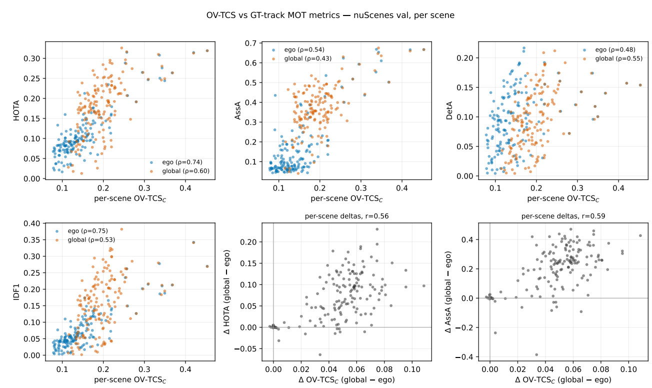

Figure 3: Per-scene OV-TCS vs. GT-based MOT metrics (nuScenes val, 150 scenes per arm). OV-TCS, computed without any ground-truth identities, tracks the association-quality axis (HOTA/AssA/IDF1) and not the detection axis.

only weakly with the detection-driven ones (DetA 0.40, MOTA 0.27, MOTP 0.23). This dissociation is the desired one: OV-TCS is a GT-free surrogate for the association/identity axis of tracking quality, not for detection quality, which mAP already covers.

5.5 Robustness to evaluator hyperparameters

OV-TCS is computed on tracks produced by an associator with hyperparameters; a reviewer should ask whether our conclusions are an artifact of one setting. We sweep the association gate, max-age, and proposal-score threshold one-at-a-time around the production point — 30 full-set replays (Tab. 3, Fig. 4).

Three findings. (1) *Rankings are invariant*: at all 10 settings, the OV-TCS ordering (Global > Fusion ≥ Ego) and the mAP ordering (Fusion > Ego = Global, an exact tie) are unchanged; no swept region reverses any conclusion in this paper. (2) *Levels are operating-point dependent*: OV-TCS CV is 0.17–0.22 vs. 0.027 for mAP; permissive gates lengthen tracks and raise L_{norm}.

	Ego	Global	Fusion
mAP CV over grid	0.027	0.027	0.027
OV-TCS CV over grid	0.220	0.167	0.220
ordering flips (mAP)	0 / 10 settings		
ordering flips (OV-TCS)	0 / 10 settings		

Table 3: Associator-sensitivity sweep (30 full val150 replays): association gate $\in \{1, 1.5, 2, 3, 4\}$ m, max-age $\in \{1, 3, 5, 10\}$, score threshold $\in \{0, 0.1, 0.3\}$, one-at-a-time around the production point. Method orderings never flip. Grid-level Spearman(mAP, OV-TCS) $= -0.018$: the two metrics respond to disjoint parameter axes.

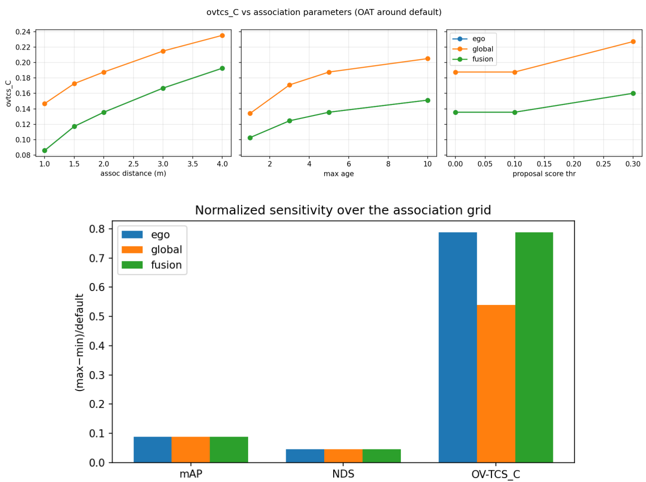

Figure 4: Associator sensitivity. Top: OV-TCS vs. each association parameter (others at default); curves shift levels but never cross — the method ordering is invariant. Bottom: normalized sensitivity per metric; association parameters move OV-TCS strongly and mAP not at all.

Absolute OV-TCS values must therefore be reported at a fixed, stated association setting; only orderings transfer across settings. (3) *Decoupling*: mAP is bit-identical across all gates and max-ages and moves only with the score threshold, while OV-TCS moves only with the gate and max-age (grid-level Spearman -0.018) — an independent, 30-point confirmation of Sec. 5.1 on real (non-injected) parameter perturbations.

6 Using the Metric

6.1 Methods mAP cannot distinguish

Table 4 is the central use case. Three pipeline decisions — association frame, IoU merging, label fusion — produce four systems whose mAP spread is 0.022 and two of which are *exactly* tied. OV-TCS and its diagnostic companions decompose the tie: global-frame association improves OV-TCS by 38% (0.136 → 0.188), cuts fragmentation by 58% (10.64 → 4.45), lengthens tracks

Method	mAP	NDS	OV-TCS	Len	Frag	CSR
Per-frame baseline	**0.3408**	**0.3150**	—[a]	—	—	—
Ego assoc.	0.3408[b]	0.3150[b]	0.136	3.01	10.64	0.507
Global assoc.	0.3408[b]	0.3150[b]	**0.188**	**3.86**	**4.45**	**0.457**
Label fusion	**0.3420**	**0.3159**	0.136[d]	3.01	10.64	0.507
M31 (IoU merge)	0.3407[e]	0.3145[e]	0.136[c]	3.01	10.64	0.507
M32 (Hungarian)	0.3198[e]	0.3028[e]	0.136[c]	3.01	10.64	0.507

Table 4: nuScenes val150 main table (closed CenterPoint stream, cache replay). OV-TCS and CSR are means over all tracks (singletons contribute 0; Sec. 3); Frag is the mean number of predicted tracks per GT instance (9,690 instances, diagnosis only). [a]class-aware baseline association pins CSR=0 structurally (degenerate). [b]association is detection-invariant: mAP/NDS unchanged. [c]merge axes act after association; track metrics equal ego by construction and reproduce it to 5 decimals ($n = 341{,}663$ tracks). [d]fusion relabels two rare classes; aggregate track metrics unchanged to the 4th decimal. [e]evaluated under an earlier revision of the detection evaluator whose anchor read 0.3407/0.3145 (offset $\leq 5{\times}10^{-4}$ from the current anchor); the merge rows' deltas are unaffected.

(3.01 → 3.86), and lowers the switch rate (0.507 → 0.457) at bit-identical detection quality. The improvement is not driven by a few scenes: the pooled (track-weighted) gain is +0.052 with 95% CI [0.049, 0.055] (scene-cluster bootstrap, 10^4 resamples; the statistic bootstrapped is the same all-track mean the metric defines), the scene-weighted mean gain is consistent at +0.049, and 96.7% of the 150 scenes improve individually. Nor is it a population artifact: the audit of Sec. 3.4 — box conservation, rising scored coverage, within-stratum gains, GT-instance-paired +0.147 — confirms the gain survives population control. Conversely, the merge and fusion axes move only detection metrics, leaving track structure untouched (verified to 5 decimals over 341,663 tracks). The evaluation cleanly factorizes into a detection axis (mAP/NDS) and a temporal axis (OV-TCS), and real design decisions live on each.

Provenance note. An earlier draft of this comparison reported global-arm OV-TCS 0.168 (+24%). That value came from a since-retired variant of the global-frame associator. The production implementation — used for every number in this paper — yields 0.188, reproduced independently by the MOT-agreement harness (Tab. 2) and the sensitivity replay (Sec. 5.5); the ego arm is bit-identical across all runs. The difference is implementation provenance, not a scientific disagreement: both variants preserve every ordering and every conclusion, and only the magnitude of the global arm's gain changes.

6.2 The effect survives an open-vocabulary label stream

The flagship comparison above uses the closed CenterPoint label stream. On a fully open-vocabulary stream (2D open-vocabulary detections lifted to 3D, 15-class prompt vocabulary), the same associator change — the same production implementations as Tab. 4 — reproduces the same signature: OV-TCS 0.216 → 0.272 (+26%), fragmentation −35% (3.76 → 2.43 predicted tracks per GT instance), switch rate over multi-frame tracks −0.041, at unchanged detection metrics (as in Tab. 4, footnote *b*: association neither adds nor removes boxes, so mAP/NDS are quoted from the shared cached evaluation of the frozen proposal set). Restricting the vocabulary to the 10 nuScenes classes leaves the deltas essentially unchanged (+25%, −35%), so the improvement is not carried by non-benchmark "stuff" classes. The effect is somewhat smaller than on the closed stream — open-vocabulary proposals are noisier and shorter-lived — but its direction and structure replicate in the regime the metric is designed for.

6.3 Class-aware label fusion

For the secondary contribution (Sec. 1), we fuse 2D open-vocabulary detections into the 3D stream with class- and source-aware gating (per-class IoU and score thresholds; here instantiated for the two classes where the 2D detector is reliably stronger, bicycle and motorcycle). Fusion raises mAP 0.3408 → 0.3420 and NDS 0.3150 → 0.3159 against the CenterPoint anchor while provably not touching track topology (Tab. 4). We report its honest boundaries: a naive global score gate for fusion *fails* (it cannot beat the anchor), and the open-vocabulary proposal stream itself remains far below the closed anchor on closed-set mAP (0.002 vs. 0.341) because its bottleneck is 3D localization, not labeling. Fusion is evidence that the 2D→3D label pathway carries exploitable signal when gated per class — not a claim of state-of-the-art detection.

7 Limitations

No identity-transfer detection. Without GT identities, OV-TCS cannot see a track that drifts from one object to another of the same class; MOT metrics remain necessary where GT tracks exist (our MOT agreement results suggest OV-TCS is a reasonable surrogate where they do not). **Granularity.** The validated granularity is per-instance/per-track; per-scene aggregation does not predict per-scene AP (p=0.61) and should not be used. **Effect size.** The beyond-length effect is significant but small (ΔR^2=0.014); OV-TCS is a topological consistency signal, not a correctness estimator. **Operating-point dependence.** Absolute values are comparable only at a fixed association setting (levels vary with gate/max-age; orderings did not flip anywhere we swept, but the sweep is one-at-a-time, not exhaustive). **Aggregation is not a method.** Feeding OV-TCS back as a control signal for temporal label aggregation (EMA weighting) did not improve detection quality in our experiments (aggregation-off beat all EMA settings); we therefore position OV-TCS strictly as a metric. **Scope of systems.** All validation uses our indoor (Mask3D-tracked) and outdoor (LiDAR-proposal) pipelines, whose absolute tracking quality is modest (AMOTA 0.03–0.16); agreement with MOT metrics in a high-performance tracking regime, and breadth across third-party streaming systems, are untested and are the most important future validations. The two domains also validate complementary properties (correctness prediction indoor; MOT agreement and sensitivity outdoor), so each individual claim is currently single-domain. **Gameability.** The score can be inflated by filtering the scored track population; Sec. 3.4 defines the joint reporting rule that makes such filtering visible, and demonstrates the corresponding population-controlled audit on our own headline comparison.

8 Conclusion

Streaming open-vocabulary 3D perception sits in an evaluation blind spot: per-frame detection metrics are structurally — and, as we measured, literally bit-for-bit — unable to see temporal degradation, while every existing temporal metric (MOTA, IDF1, HOTA, AMOTA, STQ, VPQ, TETA) requires ground-truth track identities over a closed vocabulary that this setting does not have. No amount of tuning either family closes the gap, because the gap is in their inputs, not their formulas. OV-TCS fills it from the only information the setting guarantees: the system's own tracks and labels. It passes the validations a measurement instrument owes its users — sensitivity to exactly the corruptions it claims to measure, predictive signal beyond track length, agreement with GT-based MOT metrics where those are computable, and ranking stability across a 30-run sweep of its own evaluator hyperparameters — and, used on real systems, it resolves design decisions that mAP scores as exact ties. **Future work.** The most important next step is external validation: applying OV-TCS to independent, third-party streaming open-vocabulary systems and to high-performance trackers beyond our own two pipelines. Because the metric consumes only predicted tracks and labels, it can be computed on any published system output without retraining, GT access, or code integration; showing that its rankings remain consistent with GT-based assessments across frameworks is the natural path from a validated metric to a general evaluation standard. We have not yet performed this external validation, and we flag it as such (Sec. 7).

The take-home message is simple. If a system's job is to maintain a persistent semantic map, its evaluation must contain a column that can tell a persistent map from a flickering one. mAP cannot be that column. OV-TCS can — and it costs nothing but the predictions the system already makes.

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